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Department of Electrical and Information Engineering, Universitas Gadjah Mada

Lecture 13: Working Principle, Modeling, and Control of a Buck Converter

Dr.-Ing. Yohan Fajar Sidik

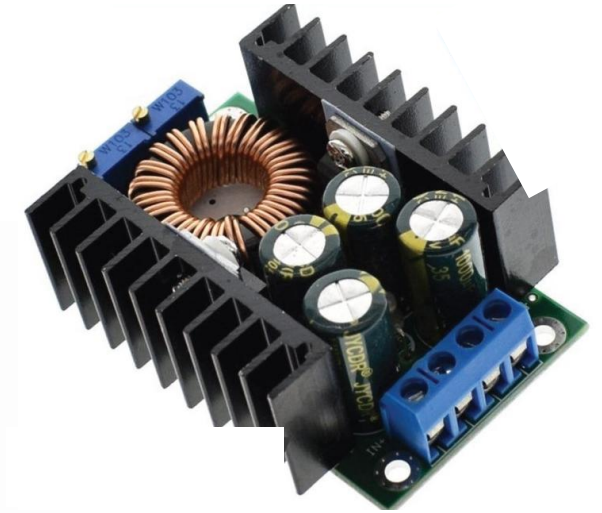
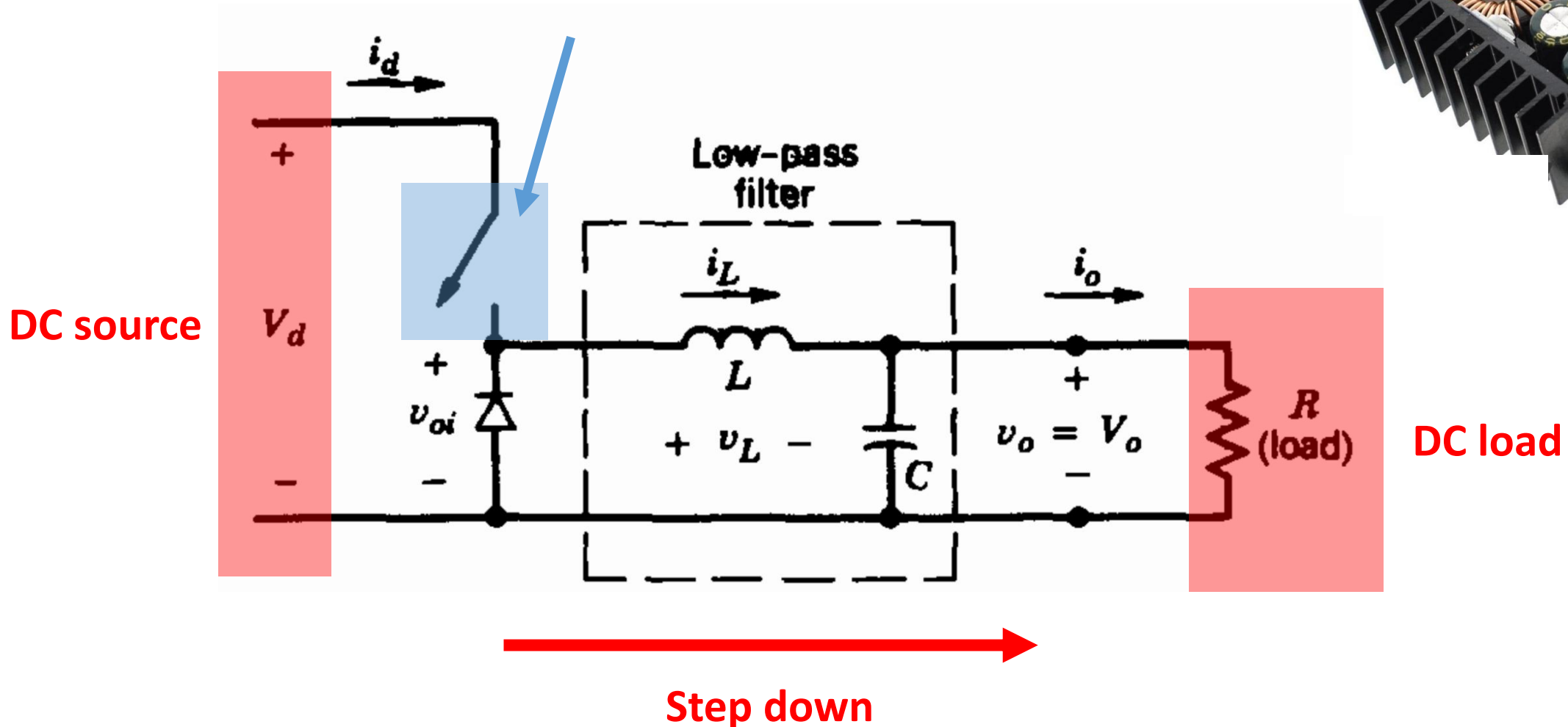
Even Semester 2024/2025

Agenda

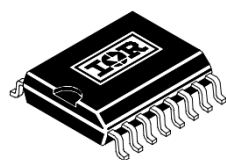
- Intro
- Working principle of a buck converter
- Modeling and control of a buck converter

Buck Converter Circuit

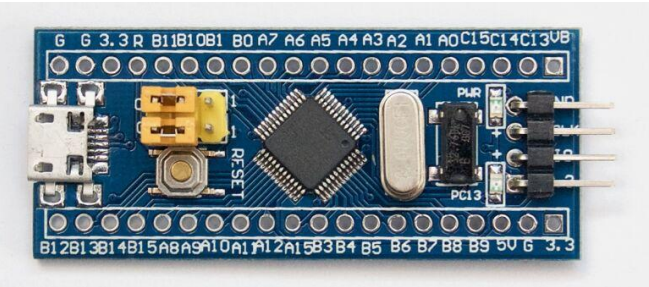
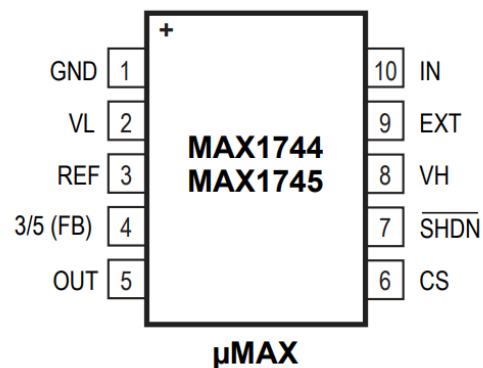
Power semiconductor device



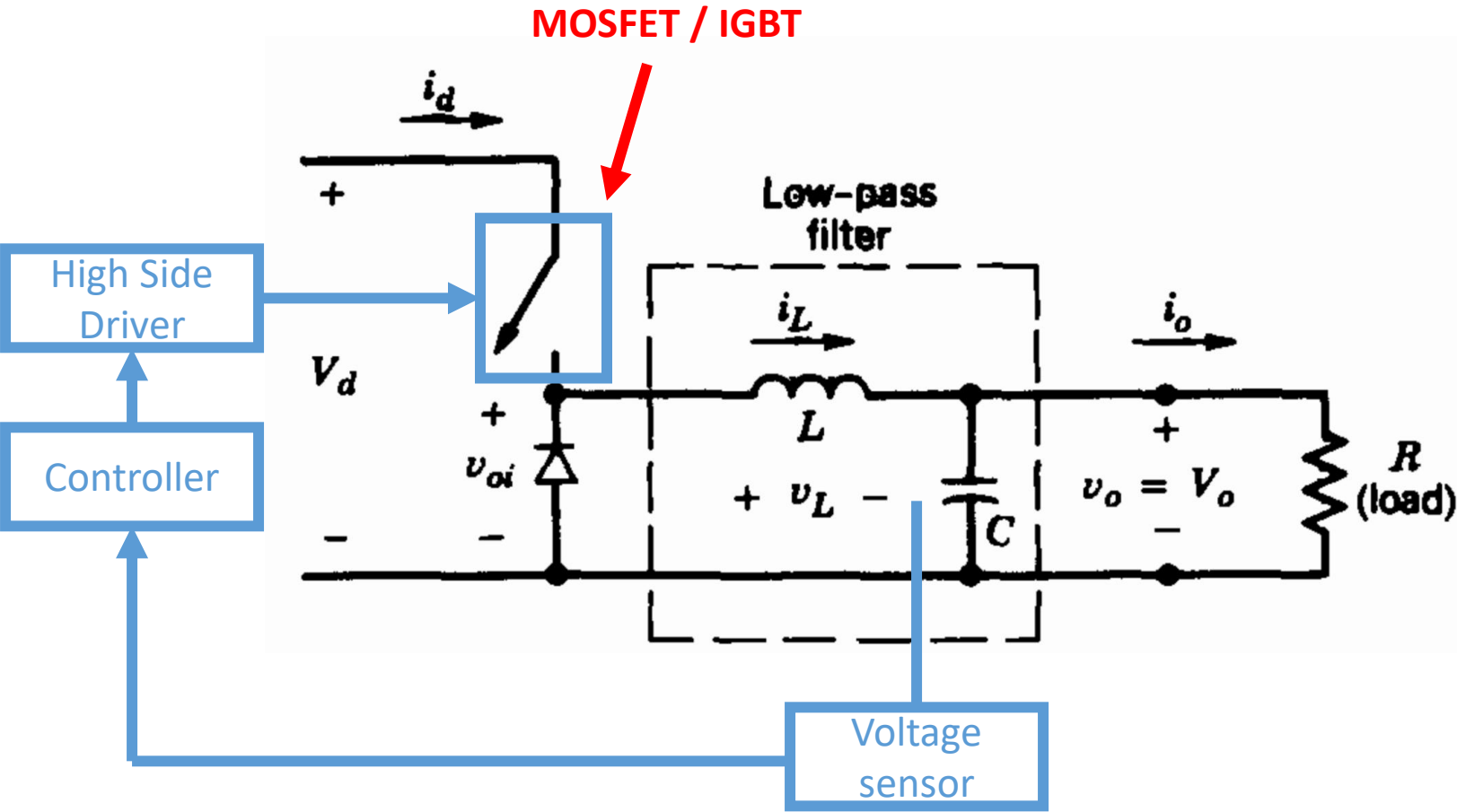
Buck Converter Components



16-Lead SOIC
IR2110S/IR2113S



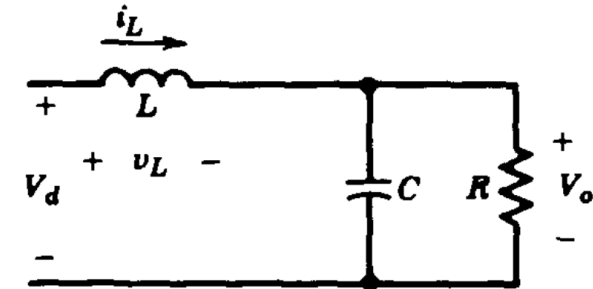
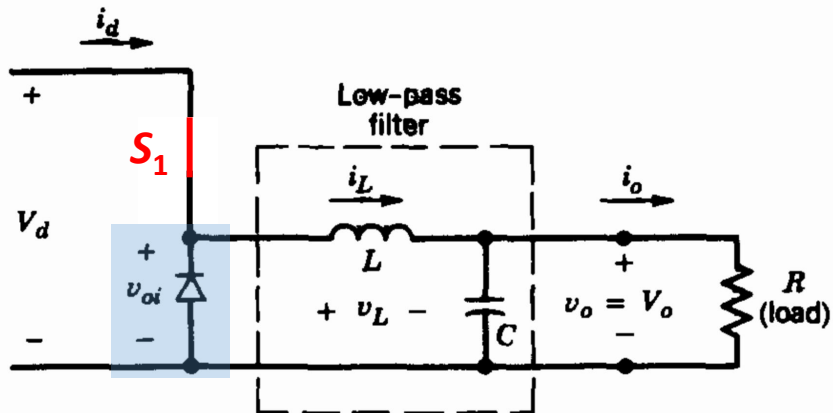
STM32 Microcontroller



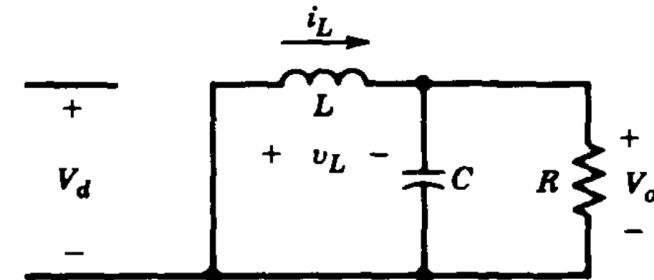
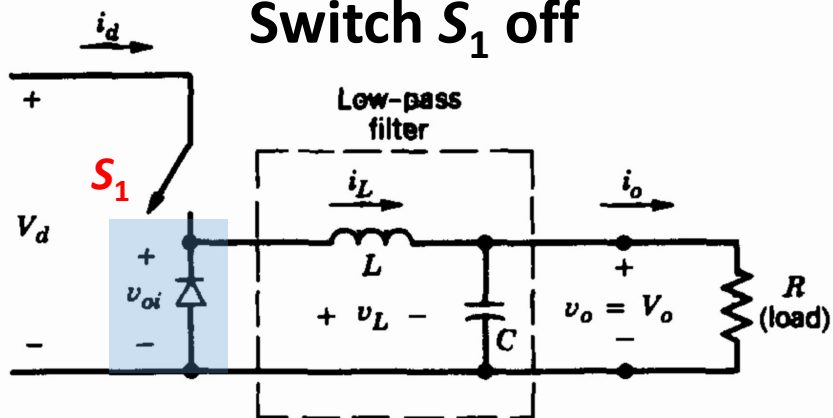
Working Principle of a Buck Converter

Working Principle of a Buck Converter

Switch S_1 on

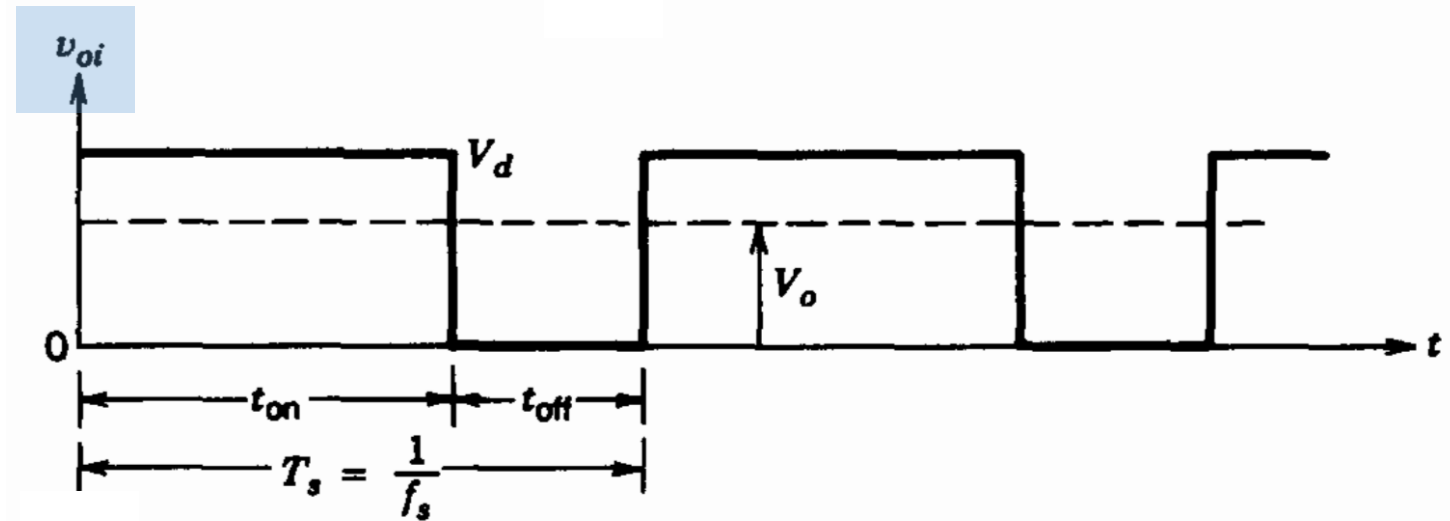
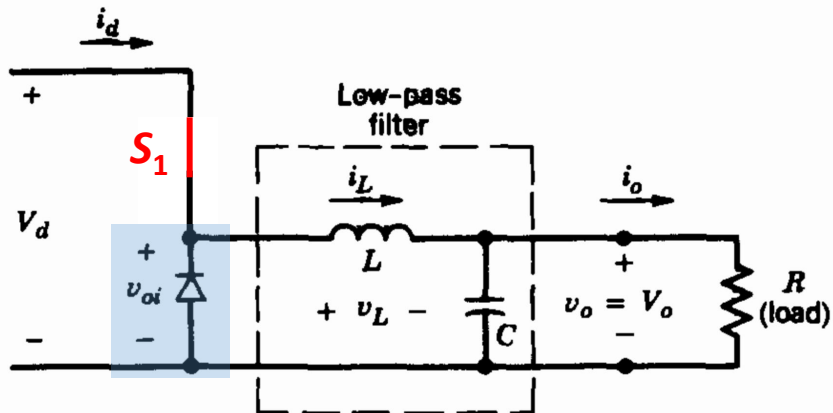


Switch S_1 off

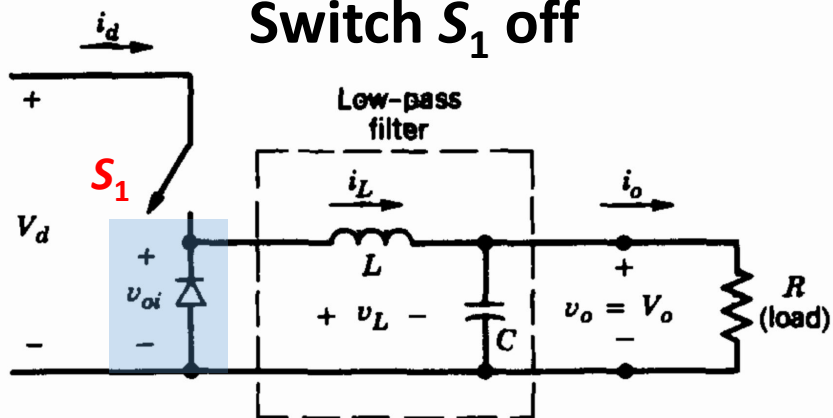


Working Principle of a Buck Converter

Switch S_1 on



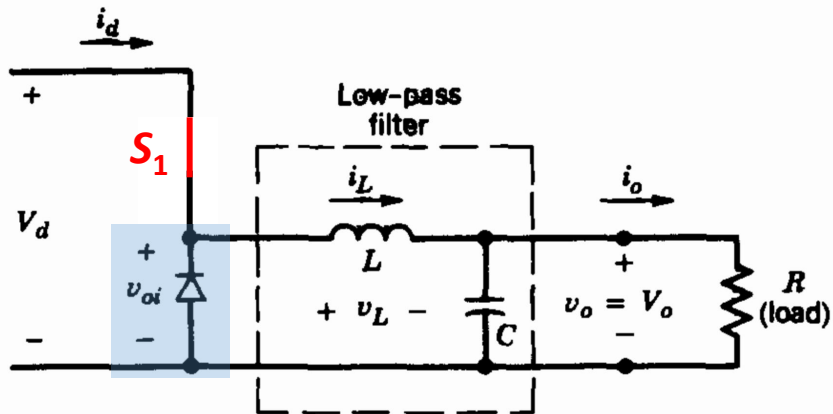
Switch S_1 off



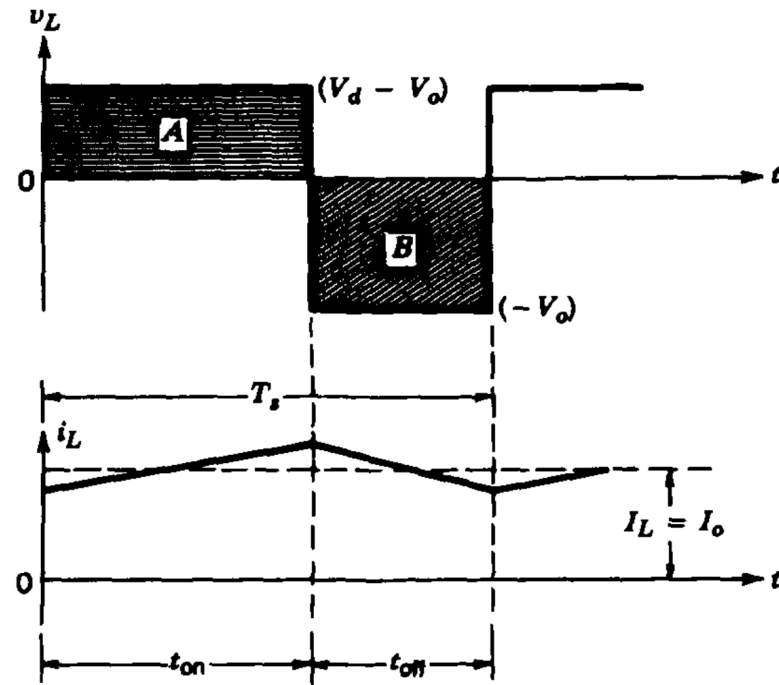
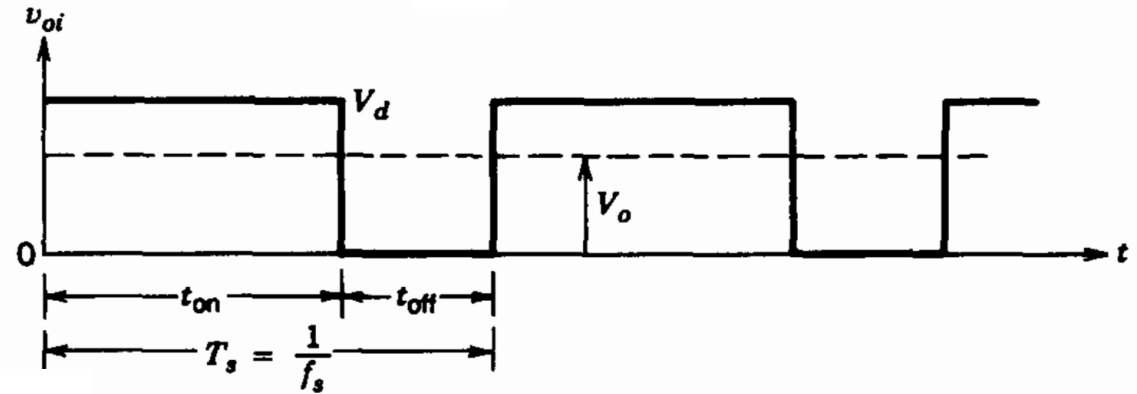
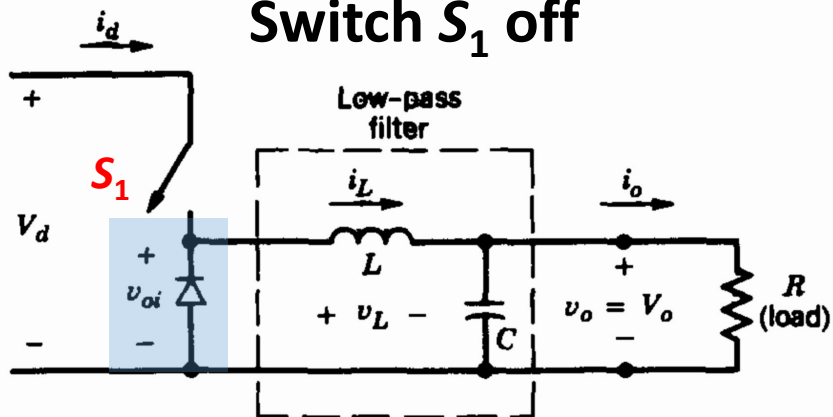
Working Principle of a Buck Converter

Continuous-Conduction Mode

Switch S_1 on

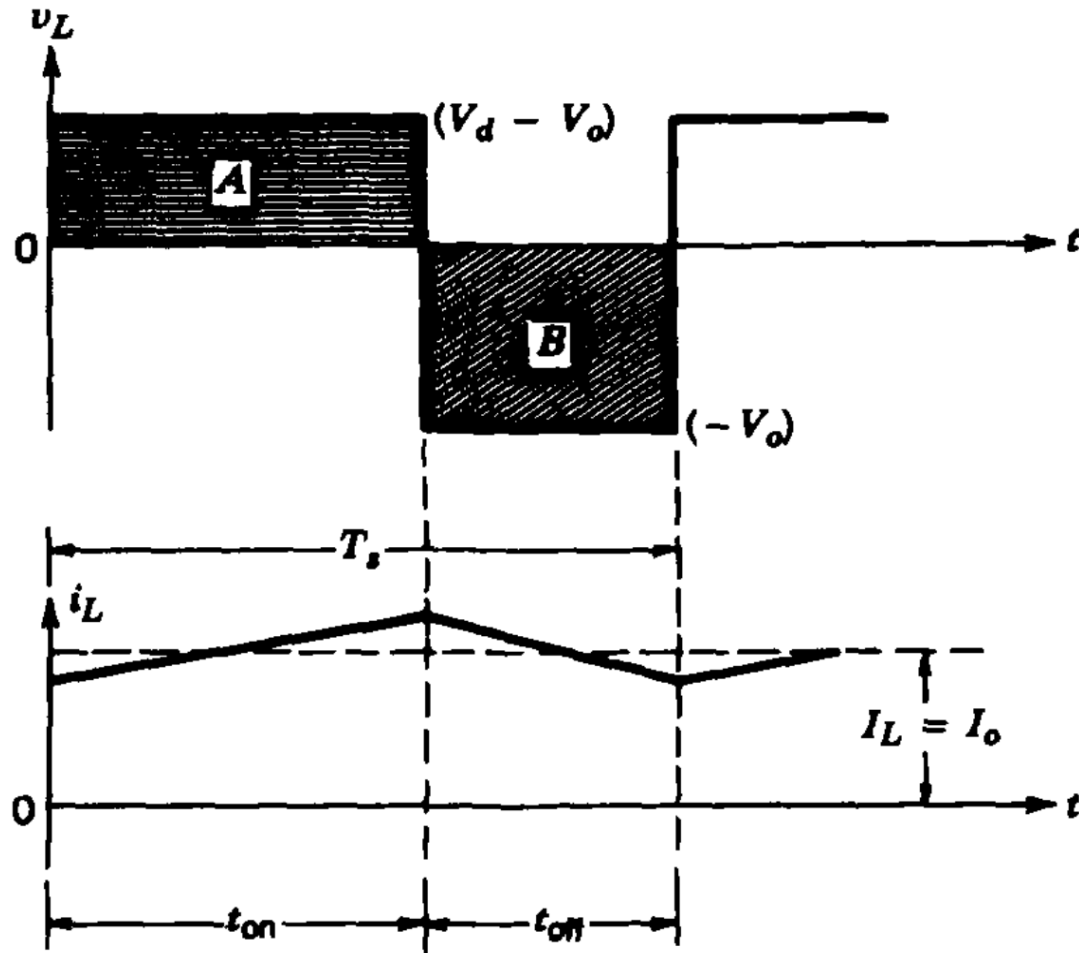


Switch S_1 off



Working Principle of a Buck Converter

Volt-Sec Balance



$$\int_0^{T_s} v_L dt = \int_0^{t_{on}} v_L dt + \int_{t_{on}}^{T_s} v_L dt = 0$$

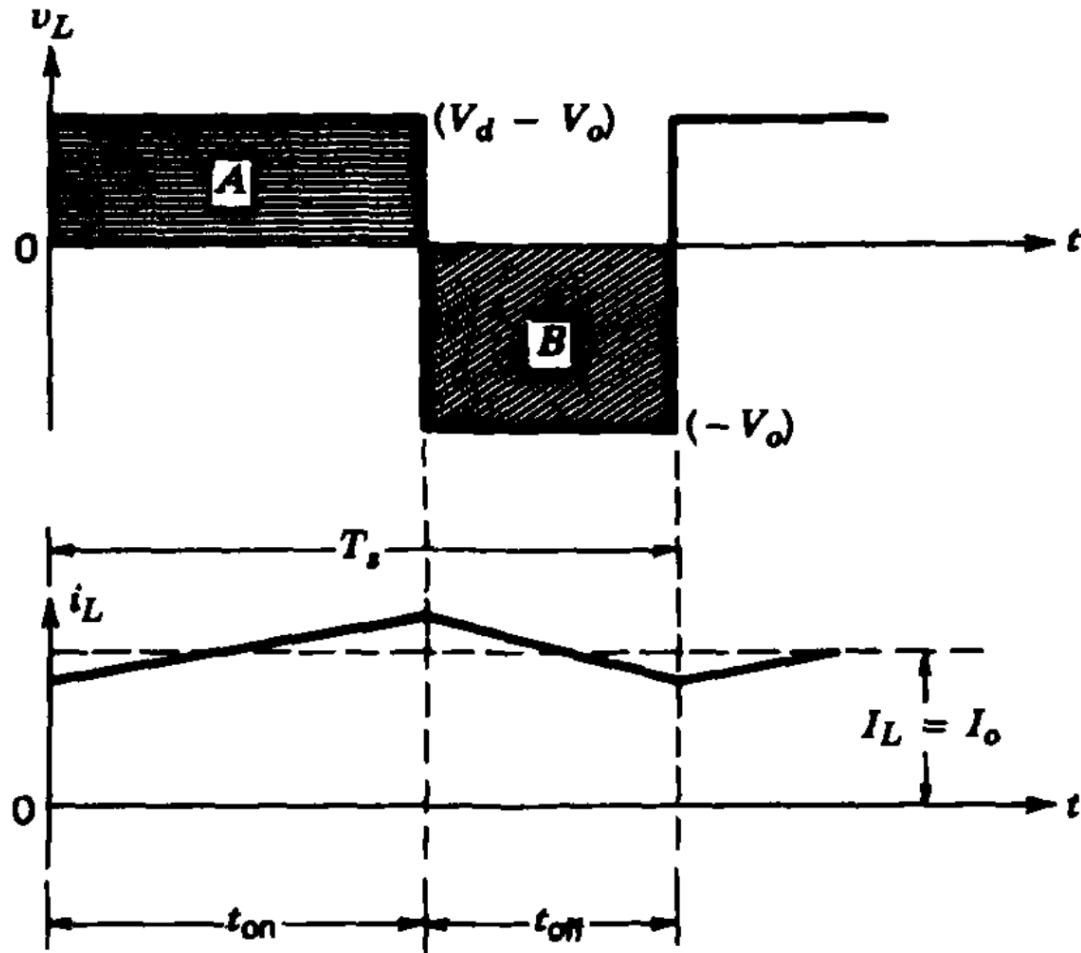
Areas A and B must be equal

$$(V_d - V_o)t_{on} = V_o(T_s - t_{on})$$

$$\frac{V_o}{V_d} = \frac{t_{on}}{T_s} = D \quad (\text{duty ratio})$$

Working Principle of a Buck Converter

Volt-Sec Balance



Neglecting power losses,

$$P_d = P_o$$

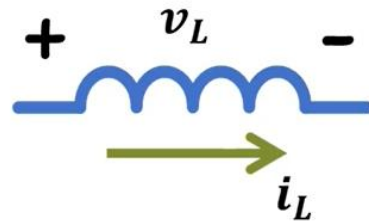
$$V_d I_d = V_o I_o$$

$$\frac{I_o}{I_d} = \frac{V_d}{V_o} = \frac{1}{D}$$

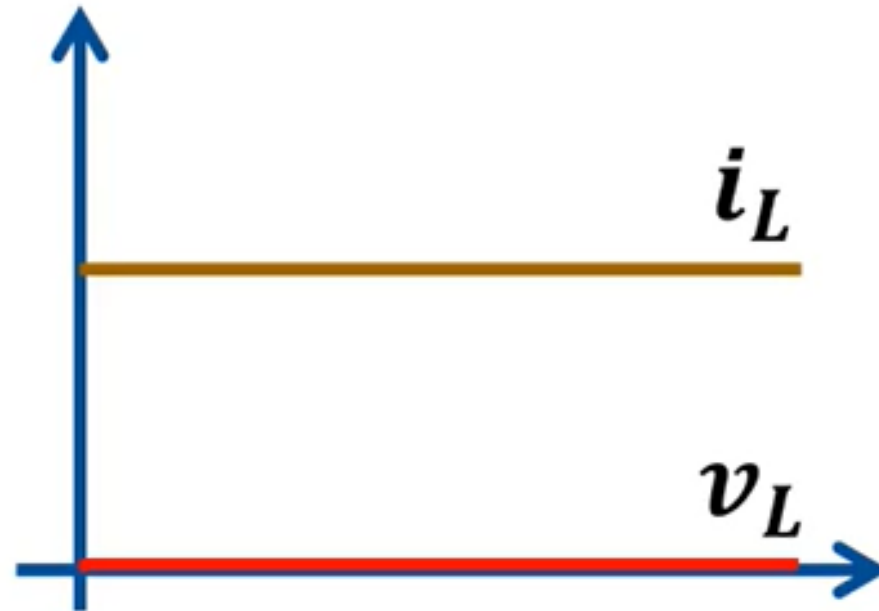
Working Principle of a Buck Converter

Basic → Volt-Sec Balance

In DC circuits, at steady state, the inductor behaves like a wire and the voltage across it is zero → $V_L = 0$



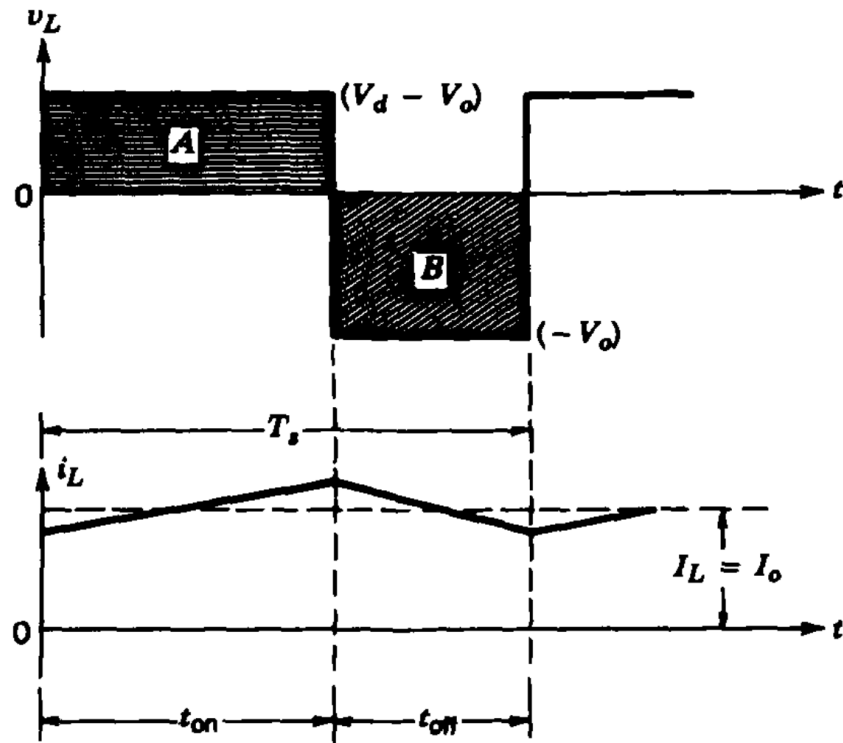
$$v_L = L \frac{di_L}{dt}$$



Working Principle of a Buck Converter

Basic → Volt-Sec Balance

The same concept is applied in switching circuits. The average voltage across an inductor over a complete switching cycle in the steady state is zero.



$$\int_0^{T_s} v_L dt = \int_0^{t_{on}} v_L dt + \int_{t_{on}}^{T_s} v_L dt = 0$$

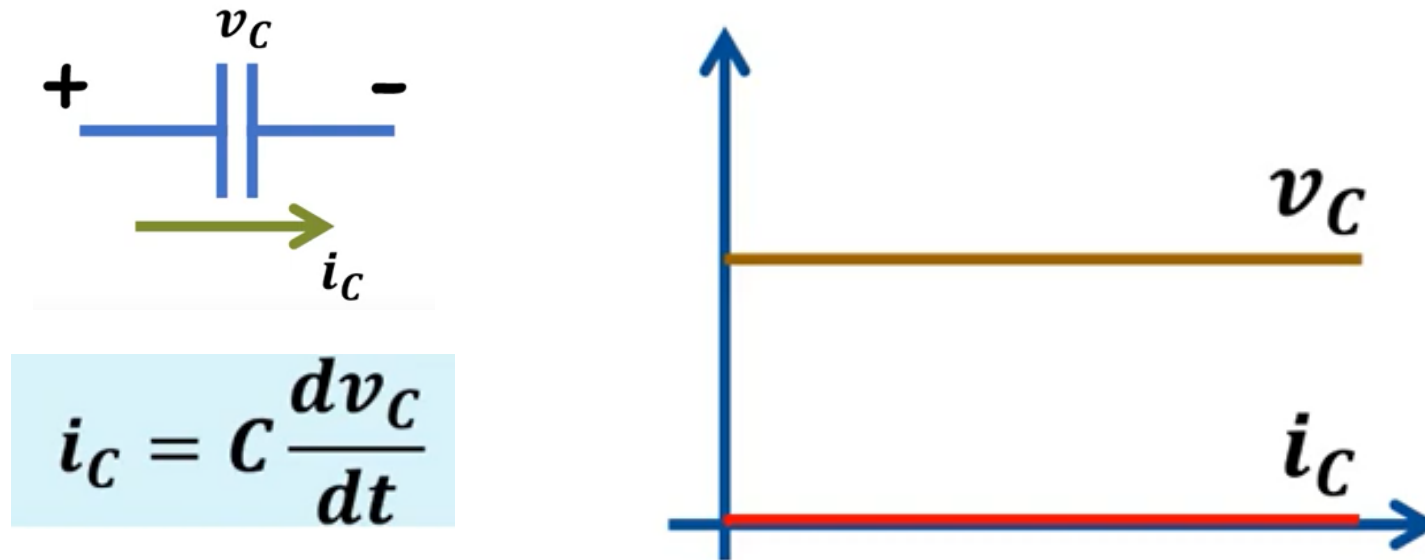
$$v_{L(avg)} = \frac{1}{T} \int_0^T v_L(t) dt = 0$$

Volt-sec balance

Working Principle of a Buck Converter

Basic → Capacitor-Charge Balance

In DC circuits, at steady state, the capacitor behaves like an open circuit and the current through it is zero → $i_c = 0$

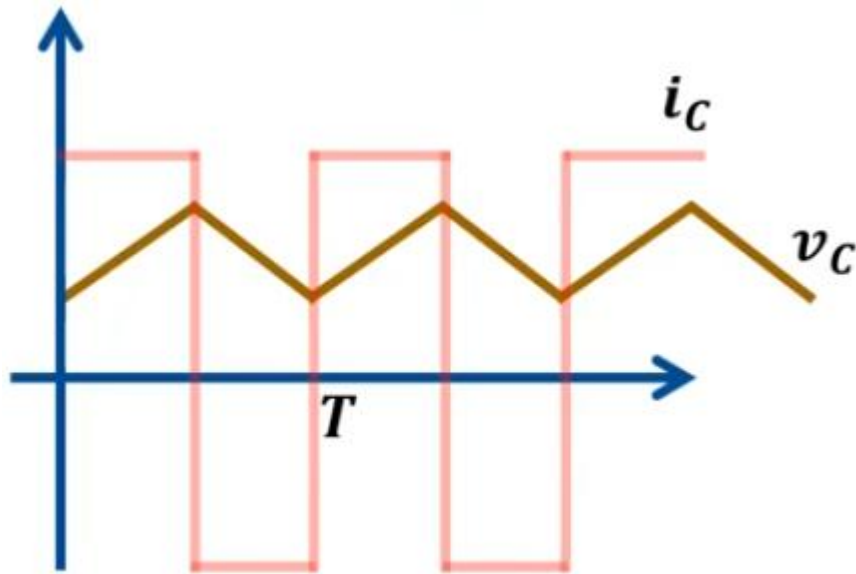


<https://www.youtube.com/watch?v=0vFbq0YkEIY>

Working Principle of a Buck Converter

Basic → Capacitor-Charge Balance

The same concept is applied in switching circuits. The average current passing a capacitor over a complete switching cycle in the steady state is zero.



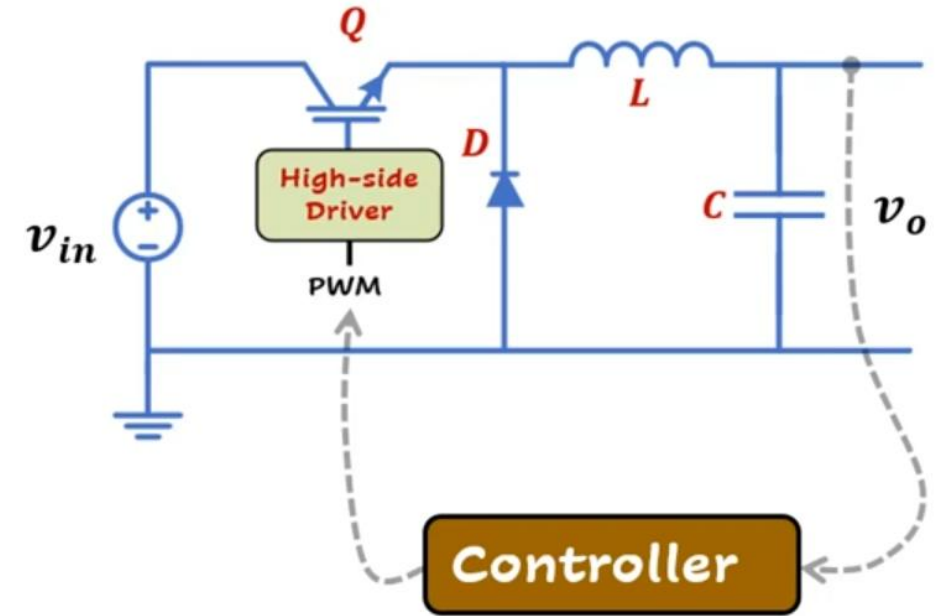
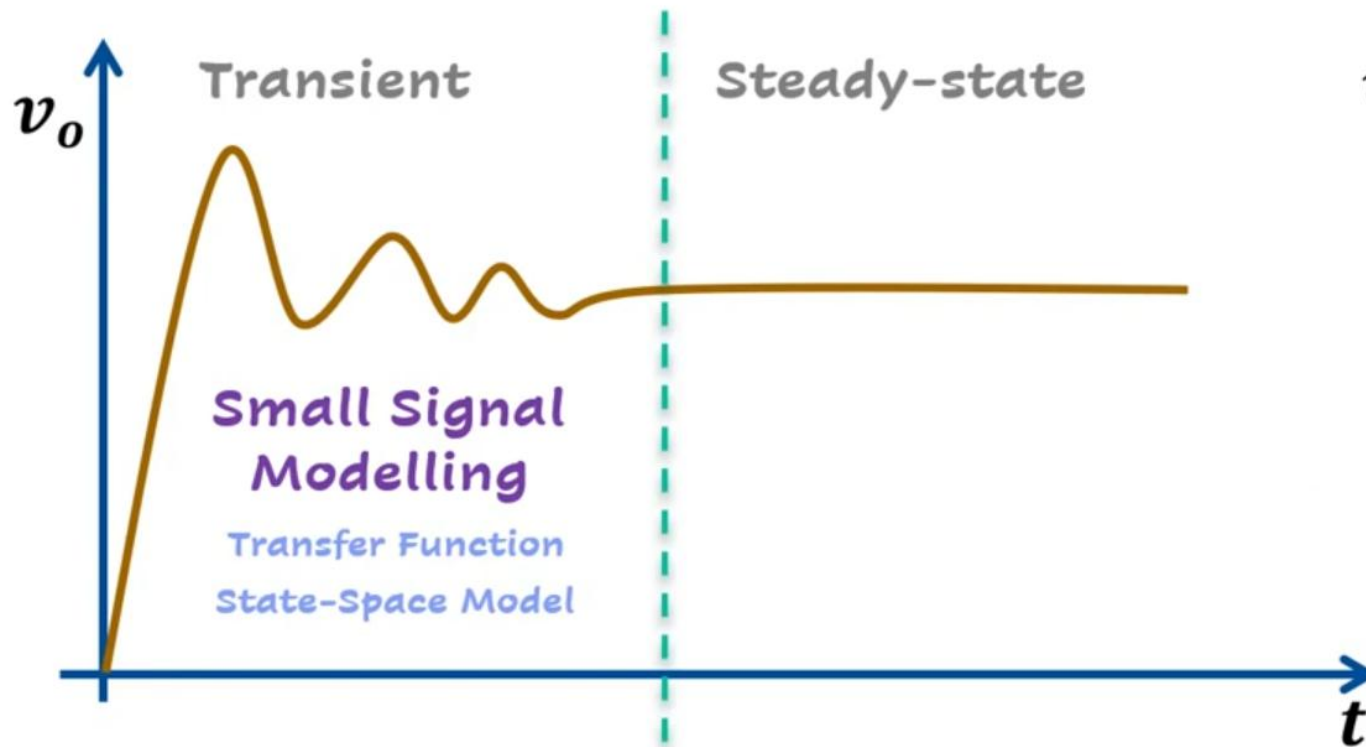
$$i_{C(avg)} = \frac{1}{T} \int_0^T i_C(t) dt = 0$$

Working Principle of a Buck Converter

Transient and Steady-State Conditions

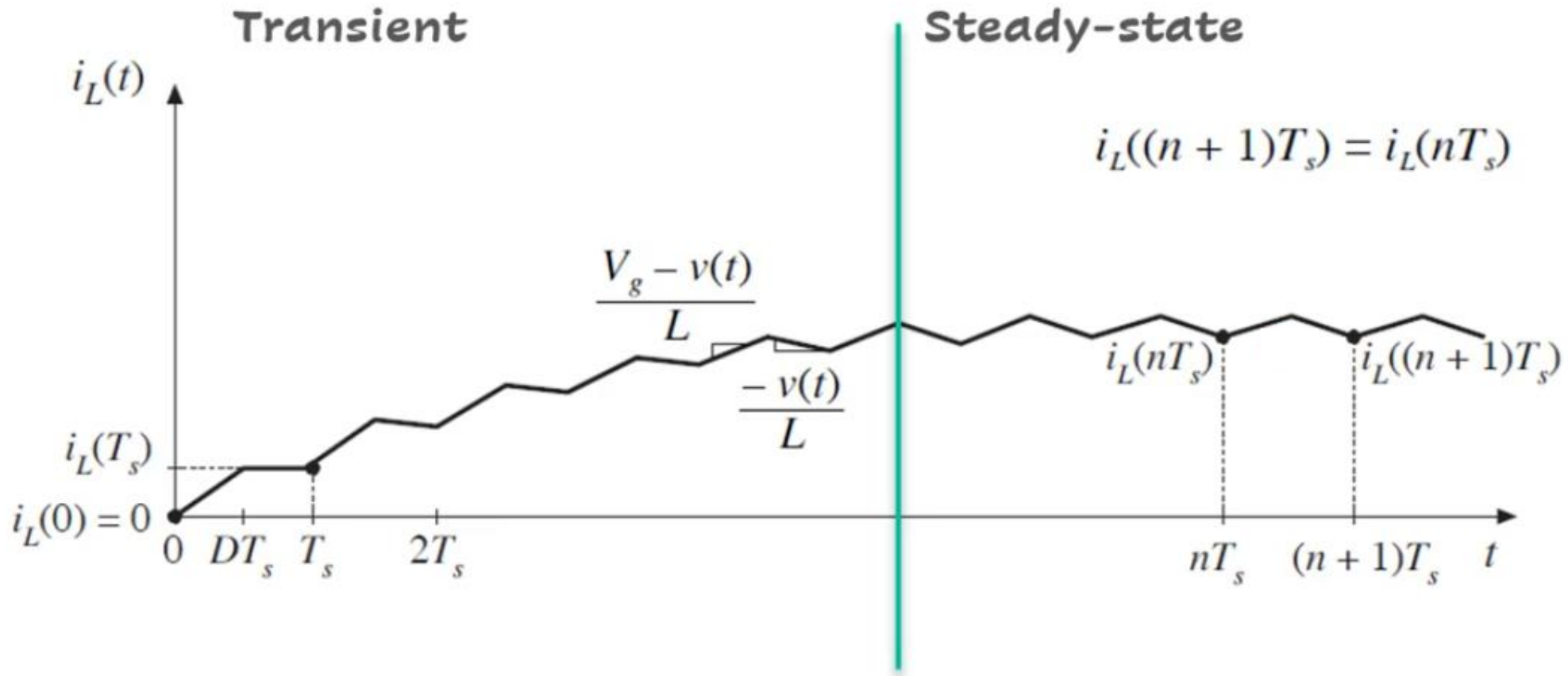
$$v_o = \delta v_{in}$$

Steady-state
input/output
equation



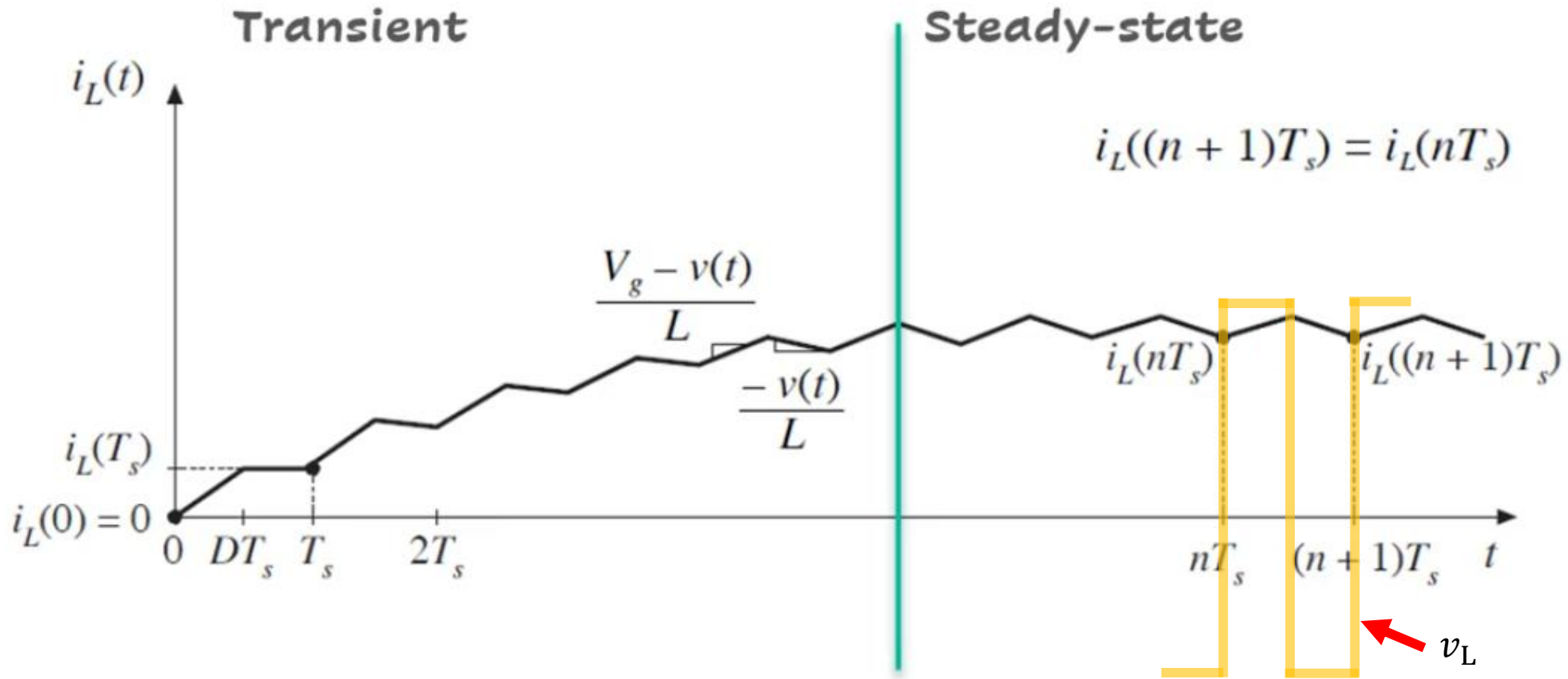
Working Principle of a Buck Converter

Transient Condition



Working Principle of a Buck Converter

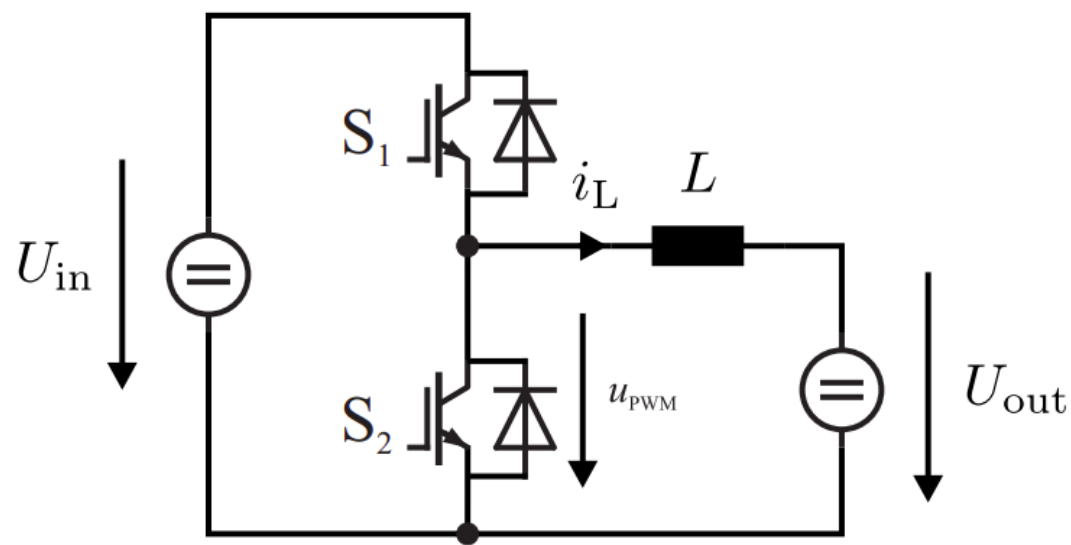
Transient Condition



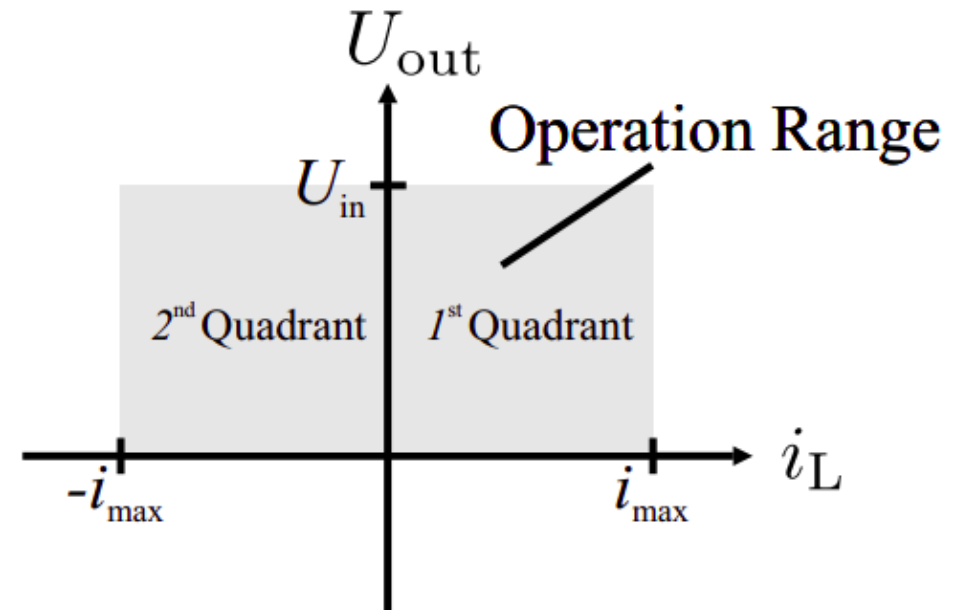
The switching circuits rely on the inductors and capacitors ability to store/release energy over the switching time

Modeling and Control of a Half-Bridge Converter (Buck Converter)

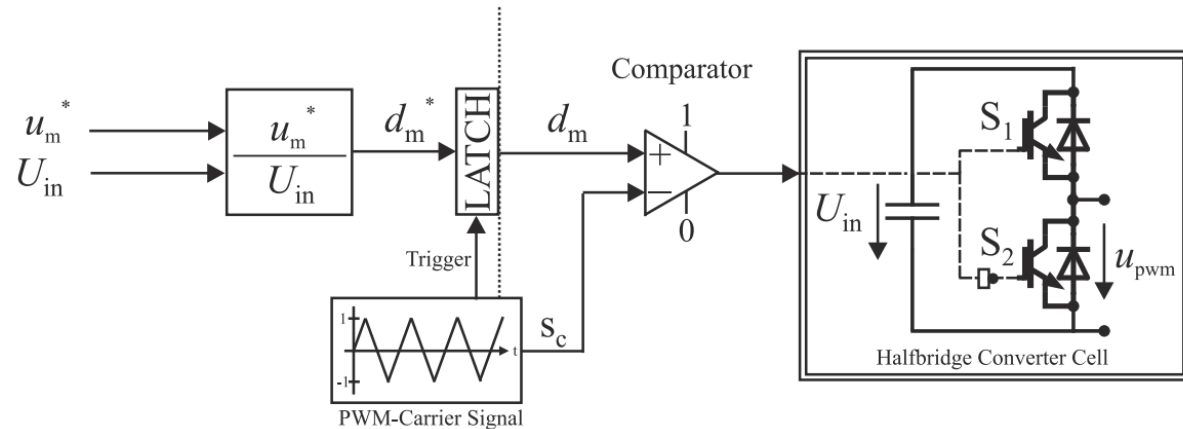
Topology and Operation Range



Half-Bridge Converter

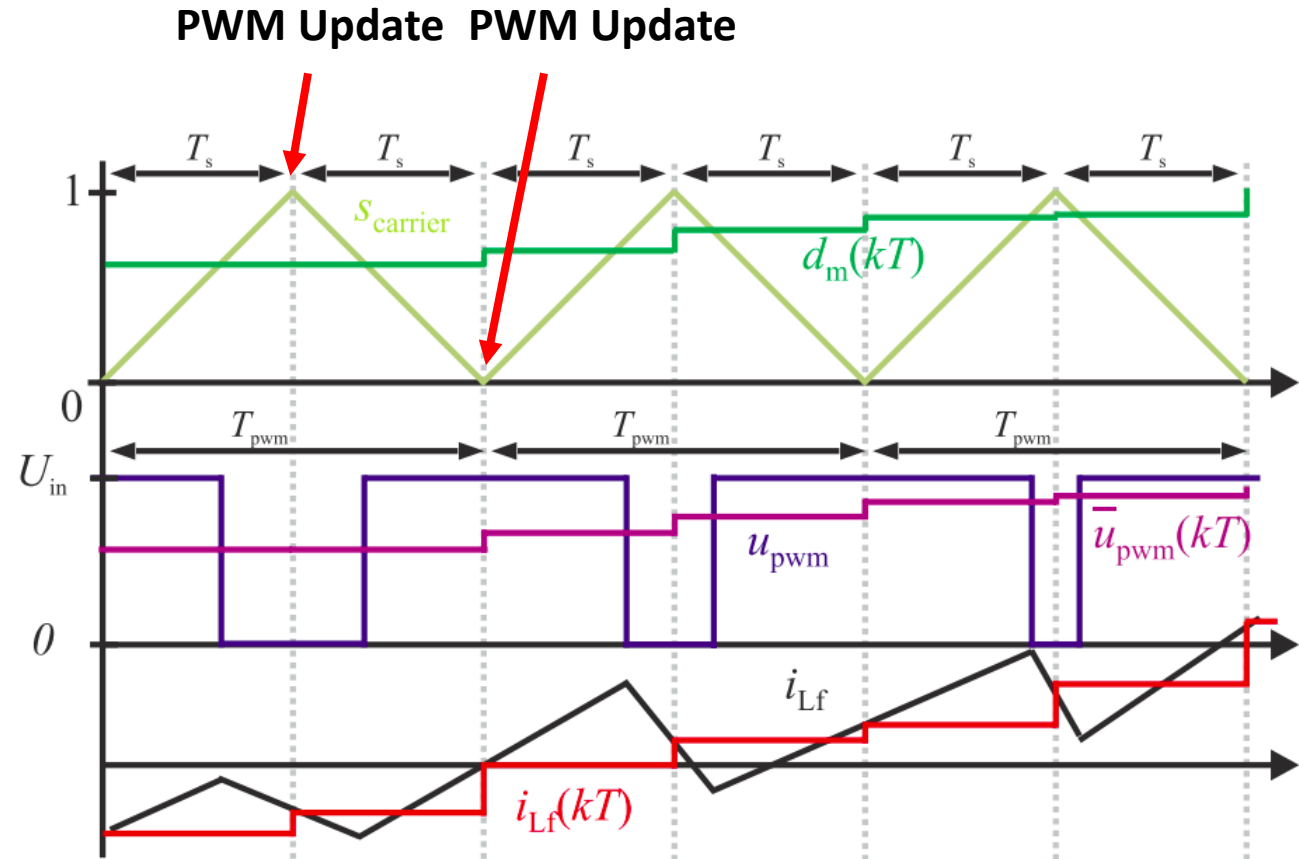


State Block Diagram of the PWM Modulation Process

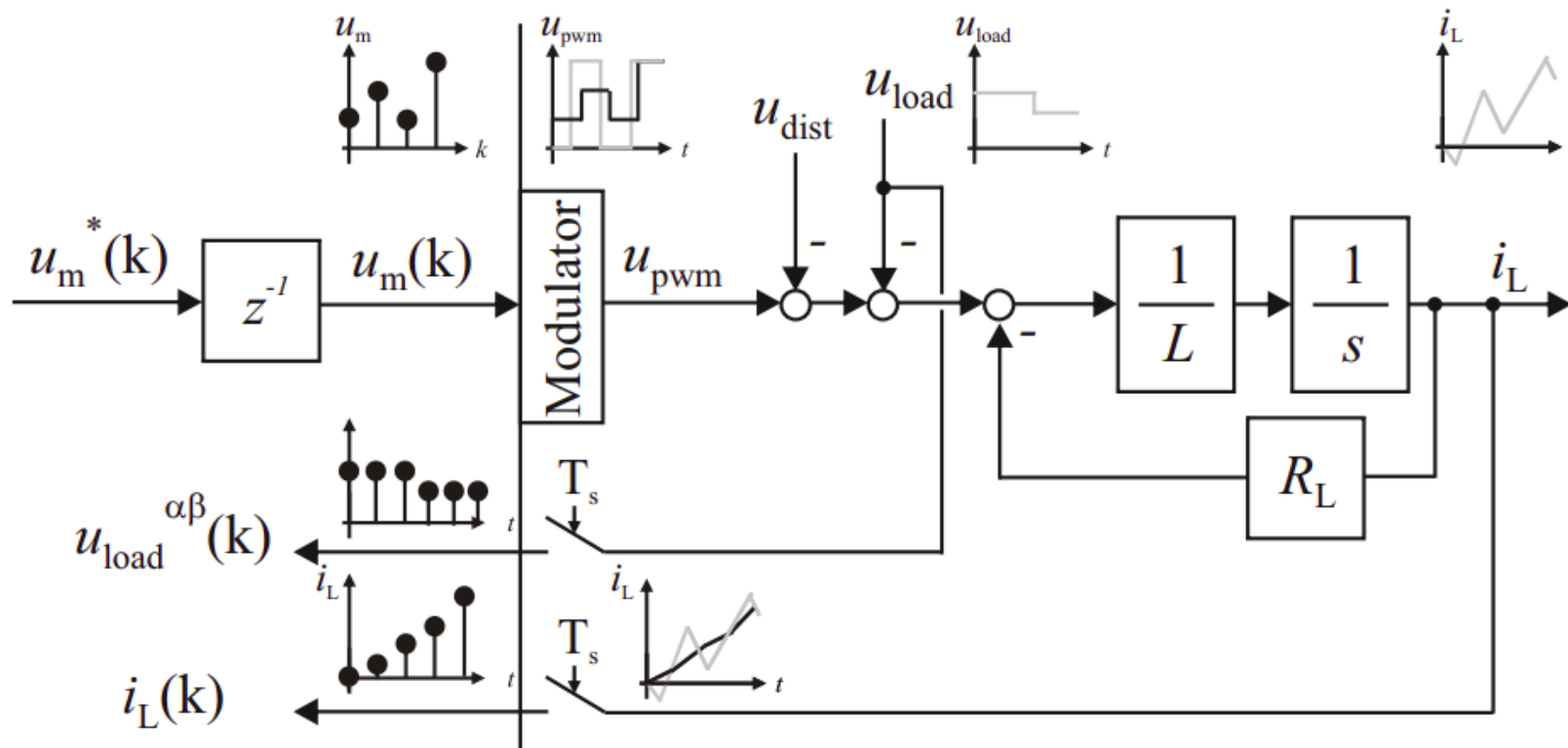


$$\bar{u}_{pwm}(kT_s) = d_m^*(kT_s) \cdot U_{in}$$

$$T_s = T_{Update} = \frac{1}{2} \cdot T_{PWM}$$

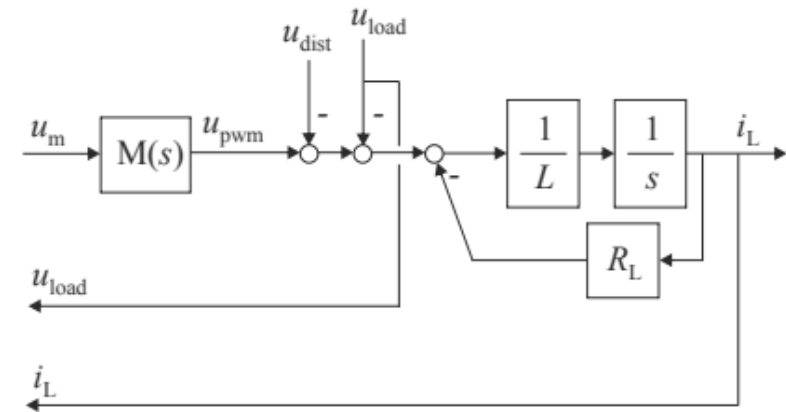
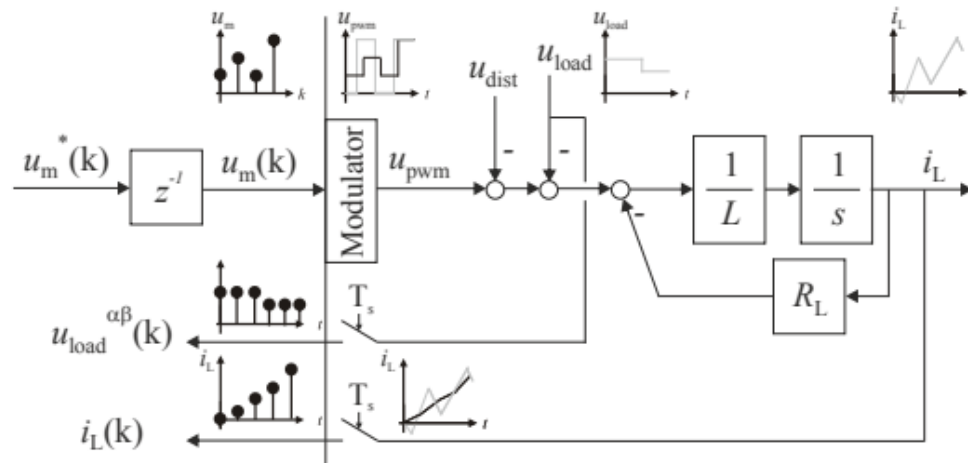
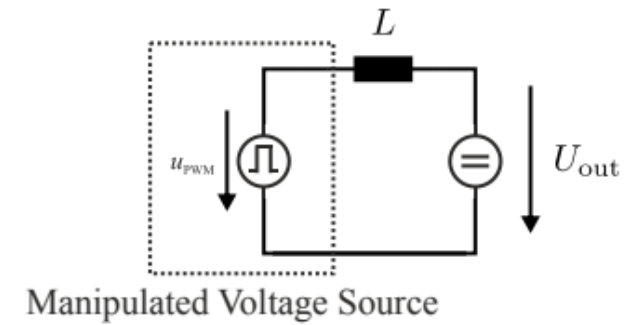
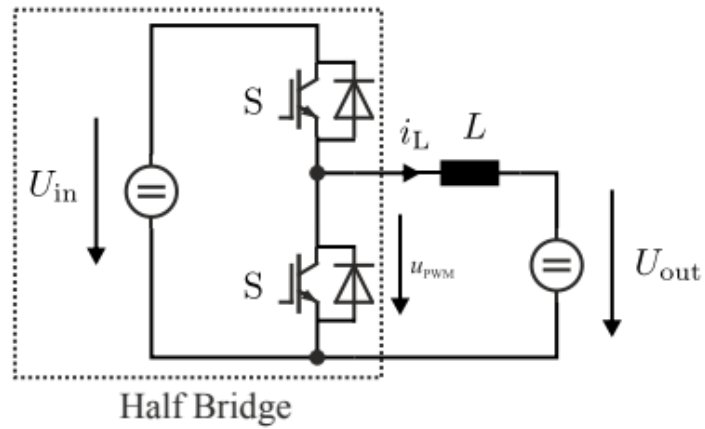


State Block Diagram with the Detailed Converter Model for Control Design

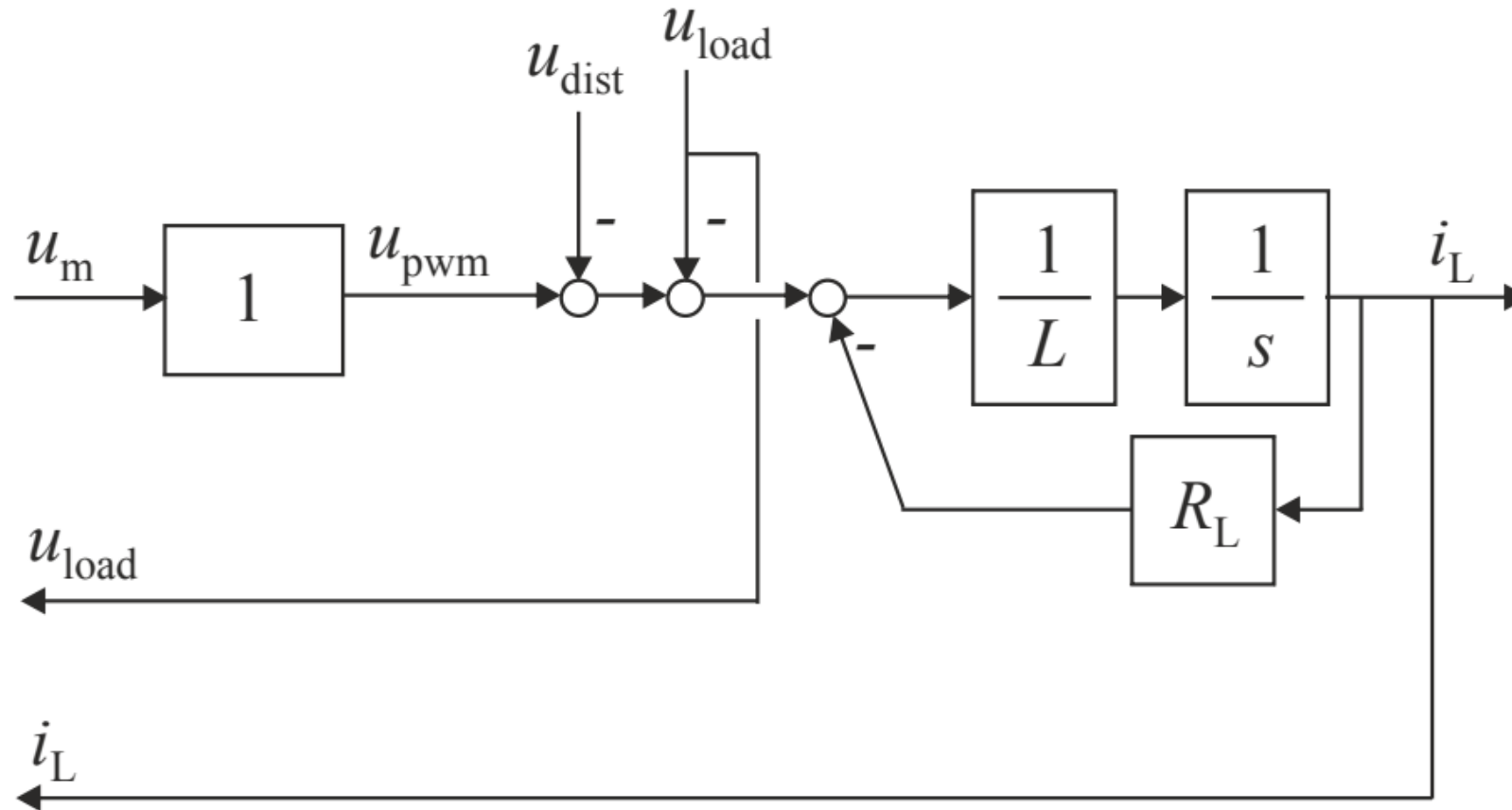


Ideal Modulator	$M(s) = 1$
Delay Model	$M(s) = e^{-s\tau}$
Low Pass Model	$M(s) = \frac{1}{s\tau + 1}$

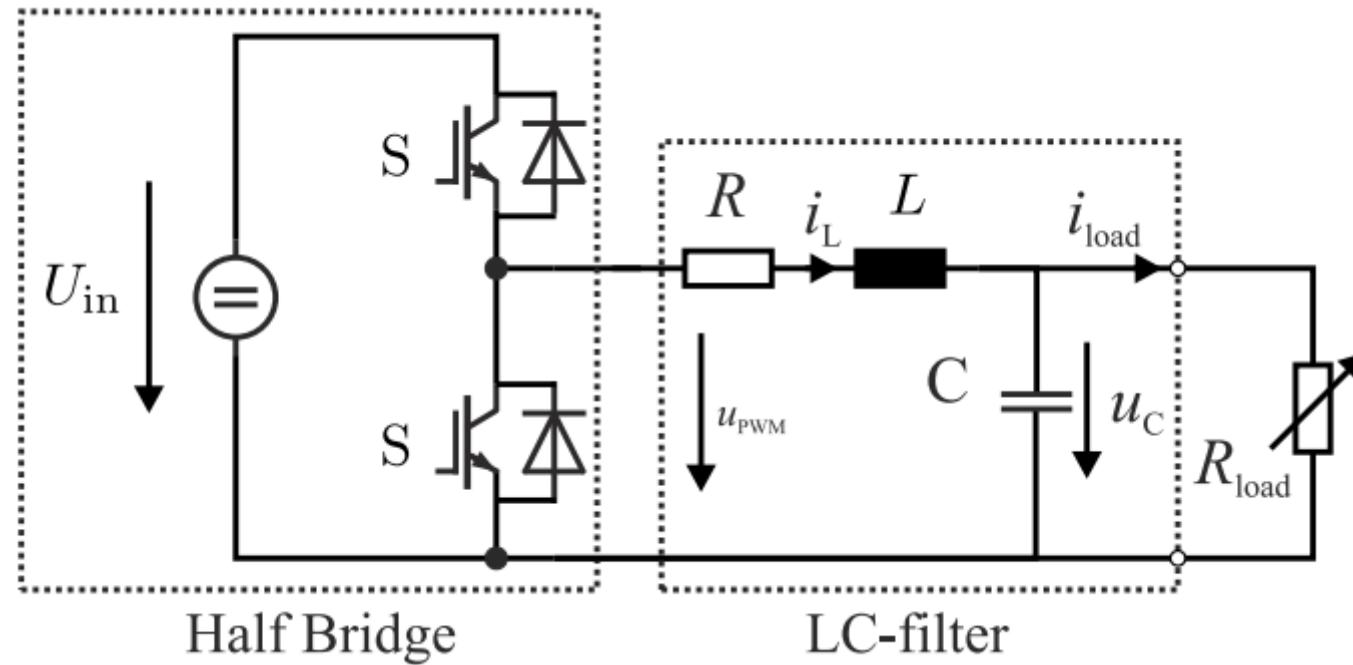
Development of a Simplified Continuous Time Model



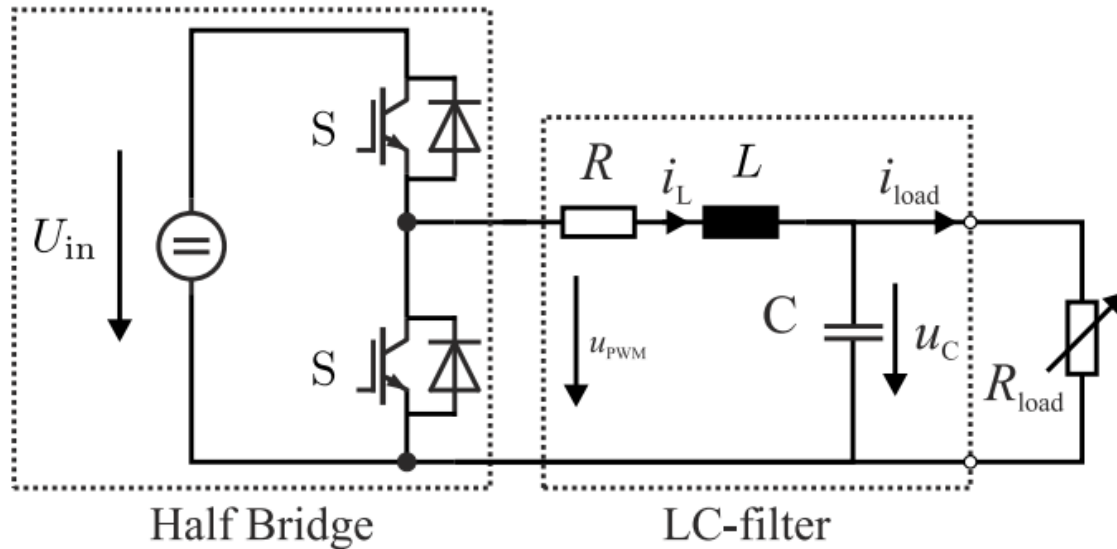
Model of the Half-Bridge Converter with Ideal Modulator $M(s)=1$



Half-Bridge Converter with LC Filter



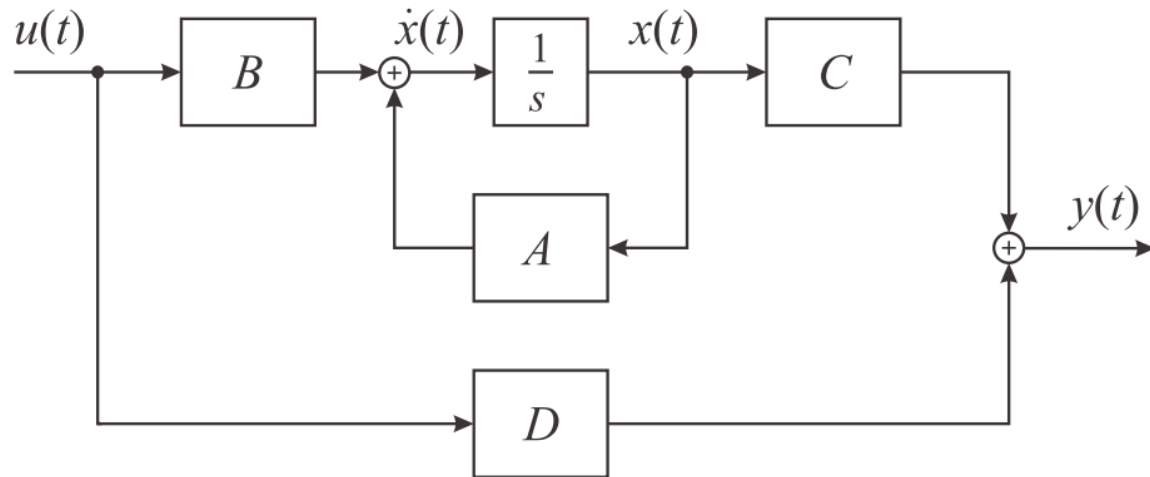
Development of the Model of Half-Bridge Converter with LC Filter



$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \\ \mathbf{y}(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t),\end{aligned}$$

State-Space Model

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$
$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t),$$



$$L \frac{di_L(t)}{dt} = u_m(t) - u_C(t) - R \cdot i_L(t)$$

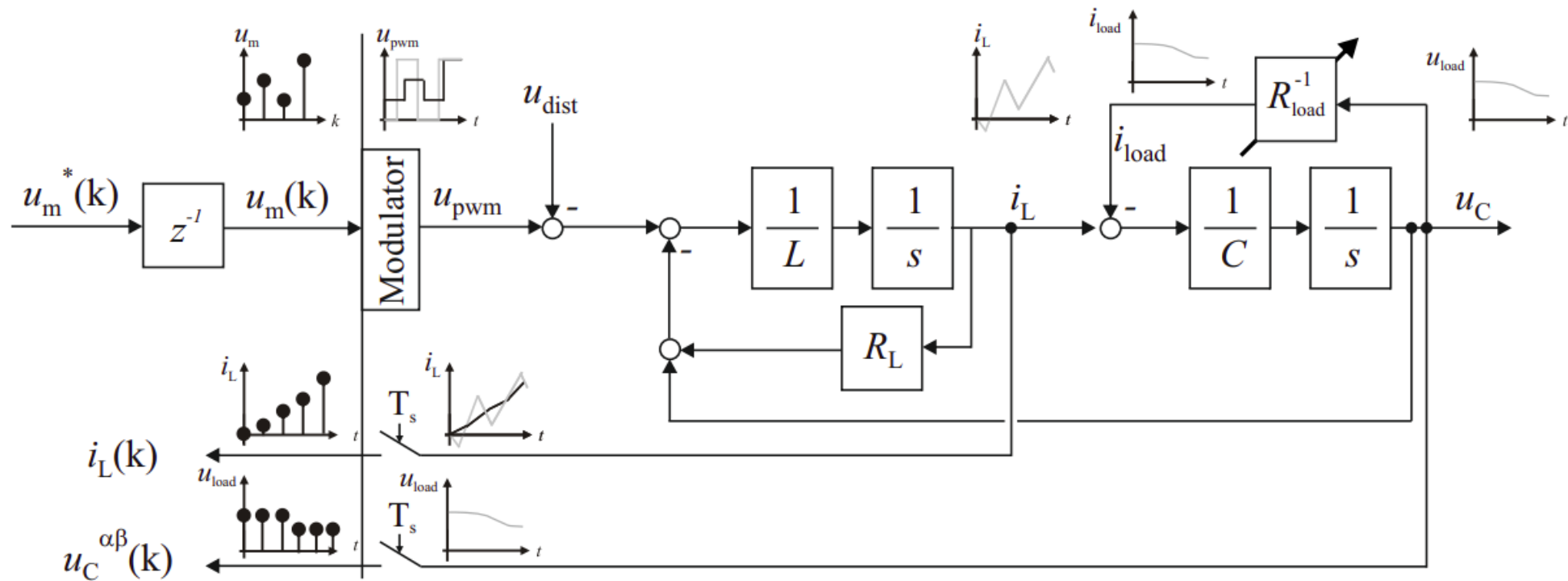
$$C \frac{du_C(t)}{dt} = i_L(t) - i_{\text{load}}(t) = i_L(t) - \frac{u_C(t)}{R_{\text{load}}}.$$

$$\mathbf{x}(t) = \begin{pmatrix} i_L(t) \\ u_C(t) \end{pmatrix} \quad \mathbf{u}(t) = (u_m(t))$$

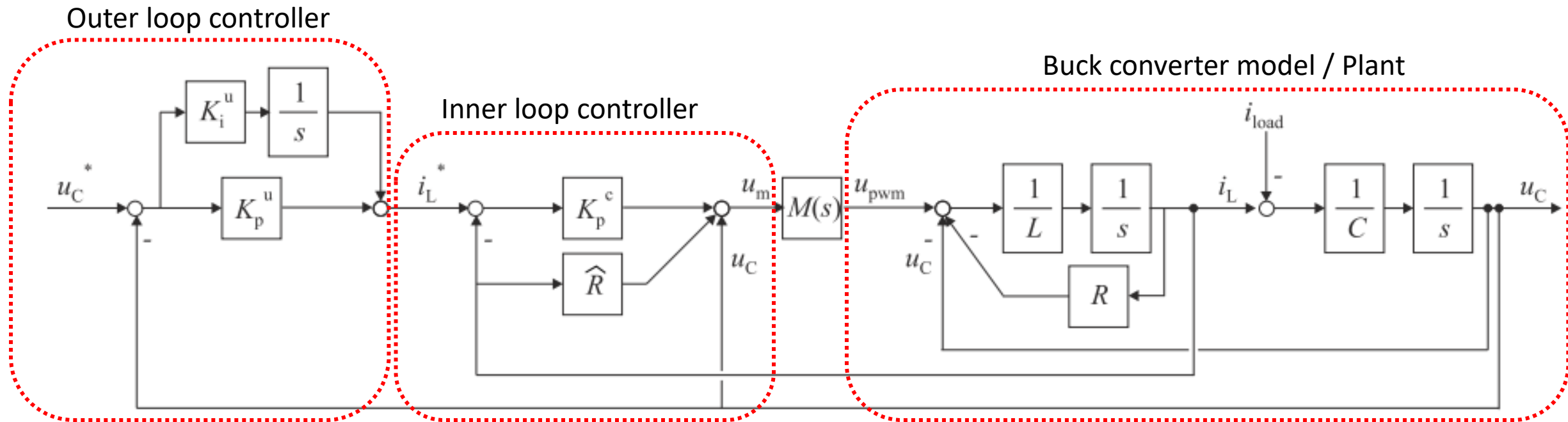
$$\mathbf{A} = \begin{pmatrix} -\frac{R}{L} & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{R_{\text{load}}C} \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} \frac{1}{L} \\ 0 \end{pmatrix}$$

$$\mathbf{C} = \begin{pmatrix} 0 & 1 \end{pmatrix} \quad \mathbf{D} = 0$$

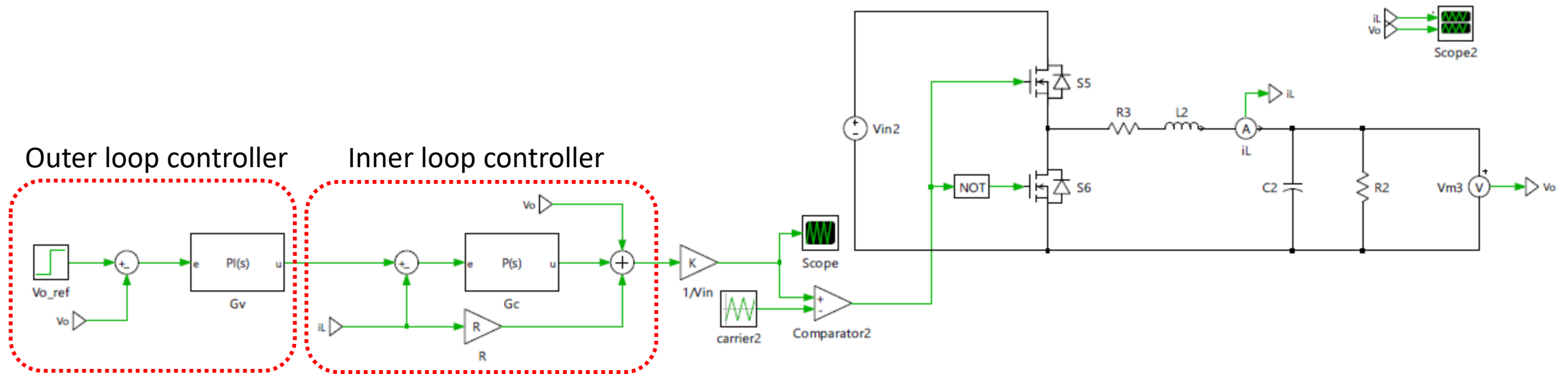
State Block Diagram of the Half-Bridge Converter with LC Filter



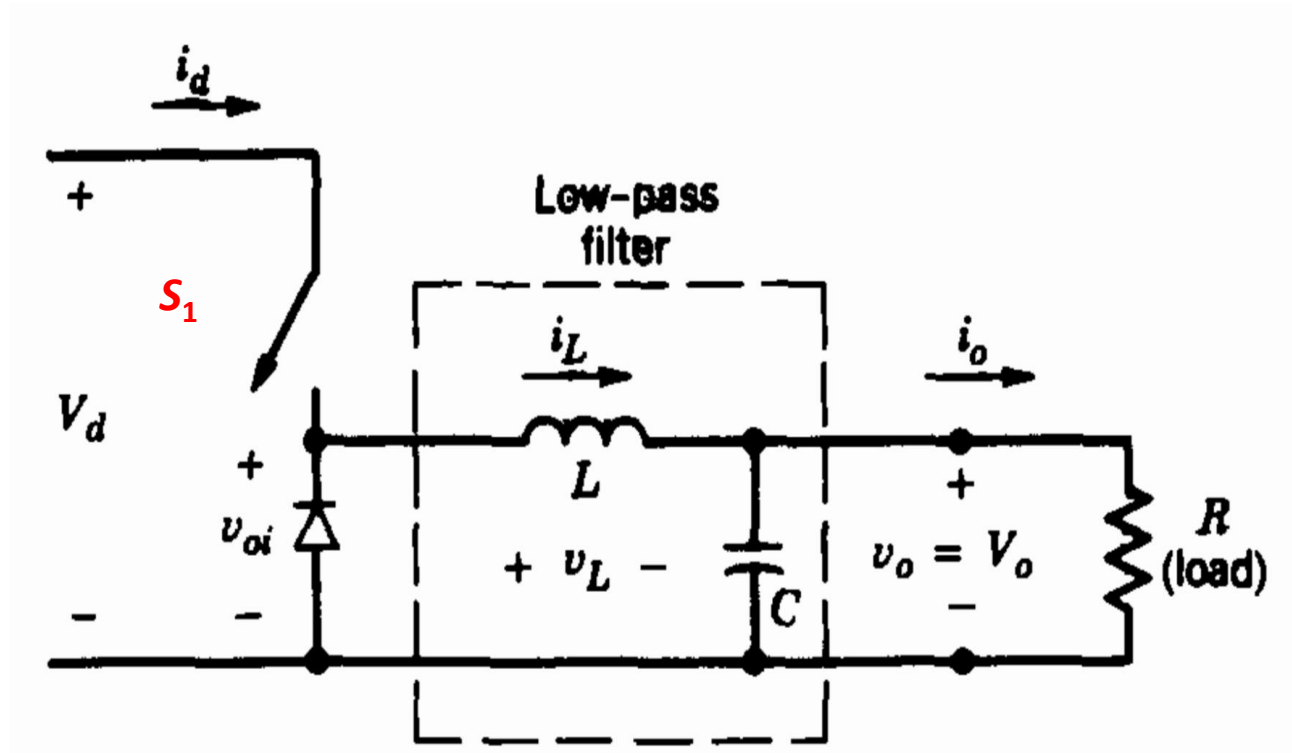
State Block Diagram of The Cascaded Control Loop



PLECS Simulation: Cascaded Control



Exercise



Converter Specifications:

$$V_{in} = 36 \text{ V}$$

$$V_o = 12 \text{ V}$$

$$f_{sw} = 100 \text{ kHz}$$

$$R_{o,min} = 1.2 \text{ ohm}$$

$$\Delta i_L = 20 \%$$

$$\Delta v_o = \pm 2 \%$$

- Calculate L and C !
- Model the buck converter circuit!
- Design a closed loop controller!

References

- Power Electronics: Converters, Applications, and Design; Ned Mohan; Wiley.
- Lecture Notes: Power Electronics: Control, Synthesis, Application; RWTH Aachen University.
- [Step-down buck converter; Imperix.](#)
- [Cascaded voltage controller; Imperix.](#)



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A detailed pencil sketch of a large, multi-story university building with many windows and columns. In the foreground, there is a street lamp with three globes, some bushes, and a small figure of a person walking. A yellow triangle points from the sketch towards the 'THANK YOU' text.

THANK YOU